

Normalization Practice & Solutions

Normalization

Practice Problem #1

Normalization Practice #1

OrderProduct Table

<u>Order No</u>	<u>Prod No</u>	<u>Cust No</u>	Name	Addr	City	St	Zip	Order Date	Promised Date	Desc	Qty Ord	Unit Price
61384	A128	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Bookcase	4	200
61384	B381	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Cabinet	2	150
61384	R210	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Table	1	500
62890	W891	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Chair	2	300
62890	A128	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Bookcase	8	200
63129	W891	1273	Cont. Designs	123 Oak	Austin	TX	78746	12/10/15	12/29/15	Chair	6	300
64000	B381	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Cabinet	1	150
64000	A128	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Bookcase	1	200

- ❖ It's already in 1NF ...but you need to understand why!
- ❖ Convert this to 2NF. How many tables do you now have?
- ❖ Convert this to 3NF. How many tables do you now have?

SOLUTION #1

Normalization Practice #1

OrderProduct Table

<u>Order No</u>	<u>Prod No</u>	<u>Cust No</u>	Name	Addr	City	St	Zip	Order Date	Promised Date	Desc	Qty Ord	Unit Price
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61384	B381	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Cabinet	2	150
61384	R210	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Table	1	500
62890	W891	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Chair	2	300
62890	A128	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Bookcase	8	200
63129	W891	1273	Cont. Designs	123 Oak	Austin	TX	78746	12/10/15	12/29/15	Chair	6	300
64000	B381	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Cabinet	1	150
64000	A128	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Bookcase	1	200

- ❖ It's already in 1NF ...but you need to understand why!
 - **No multi-valued attributes!**
- ❖ Convert this to 2NF. How many tables do you now have?
- ❖ Convert this to 3NF. How many tables do you now have?

Resolving to 2NF

Convert the original table (which already was in 1st Normal Form) to 2nd Normal Form

OrderProduct (OrderNo, ProdNo, CustNo, Name, Addr, City, St, Zip, OrderDate, PromisedDate, Desc, QtyOrd, UnitPrice)

2NF SOLUTION:

OrderNo, ProdNo □ QtyOrd

OrderNo □ CustNo, Name, Addr, City, St, Zip, OrderDate, PromisedDate

ProdNo □ Desc, UnitPrice

Example A: 2NF

2NF SOLUTION:

OrderNo, ProdNo □ QtyOrd

OrderNo □ CustNo, Name, Addr, City, St, Zip, OrderDate, PromisedDate

ProdNo □ Desc, UnitPrice

OrderProduct Table

<u>Order No</u>	<u>Prod No</u>	Qty Ord
61384	A128	4
61384	B381	2
61384	R210	1
62890	W891	2
62890	A128	8
63129	W891	6
64000	B381	1
64000	A128	1

Order Table

<u>Order No</u>	Cust No	Name	Addr	City	St	Zip	Order Date	Promised Date
61384	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15
62890	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15
63129	1273	Cont. Designs	123 Oak	Austin	TX	78746	12/10/15	12/29/15
64000	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15

Product Table

<u>Prod No</u>	Desc	Unit Price
A128	Bookcase	200
B381	Cabinet	150
R210	Table	500
W891	Chair	300

Resolving to 3NF

Convert the 2nd Normal Form tables into 3rd Normal Form

OrderProduct (OrderNo, ProdNo, QtyOrd) ✓ Already in 3NF

Order (OrderNo, CustNo, Name, Addr, City, St, Zip, OrderDate, PromisedDate) ✓ ✓ ✓
□ □ □ □ □ □ □ □

Product (ProdNo, Desc, UnitPrice) ✓ Already in 3NF

3NF SOLUTION:

OrderNo, ProdNo □ QtyOrd

OrderNo □ CustNo, OrderDate, PromisedDate

ProdNo □ Desc, UnitPrice

CustNo □ Name, Addr, City, St, Zip
□ □

Zip □ St

Example A: 3NF

OrderProduct Table

<u>Order No</u>	<u>Prod No</u>	Qty Ord
61384	A128	4
61384	B381	2
61384	R210	1
62890	W891	2
62890	A128	8
63129	W891	6
64000	B381	1
64000	A128	1

Order Table

<u>Order No</u>	Cust No	Order Date	Promised Date
61384	1273	11/04/15	11/21/15
62890	3891	11/15/15	11/21/15
63129	1273	12/10/15	12/29/15
64000	4545	04/10/15	05/17/15

Product Table

<u>Prod No</u>	Desc	Unit Price
A128	Bookcase	200
B381	Cabinet	150
R210	Table	500
W891	Chair	300

Customer Table

<u>Cust No</u>	Name	Addr	City	Zip
1273	Cont. Designs	123 Oak	Austin	78746
3891	J Consultants	523 Pine	Waco	76712
4545	ABC Ins Co	555 Elm	OKC	73134

Zip Table

<u>Zip</u>	St
73134	OK
76712	TX
78746	TX

Practice Problem #2

Normalization Practice #2

Appointment Table

<u>Appt No</u>	Appt Date	Appt Time	Planned Duration	Appt Type	Patient ID	First Nm	Last Nm	Phone	Doctor ID	Doctor Nm
1	12/1/2015	3:00 AM	1.00	Physical	466927	Lisa	Garcia	562-3456	C678	Chapman
2	12/1/2015	3:00 AM	0.25	Shot	456789	Sue	Carey	432-1234	A528	Lopez
3	12/1/2015	3:15 AM	0.50	Flu	194756	Brandon	Pierre	432-7877	S626	Smith
4	12/2/2015	10:00 AM	0.50	Migraine	329657	Marcus	Schwartz	239-5502	A528	Lopez
5	12/2/2015	10:15 AM	0.25	Shot	987453	Mike	Jones	456-0202	G123	Gray
6	12/2/2015	10:30 AM	0.25	Shot	384788	Tonya	Johnson	432-8806	S626	Smith
7	12/2/2015	10:45 AM	0.50	Flu	438754	Iliana	Hnatt	823-4303	C678	Chapman
8	12/2/2015	11:00 AM	1.00	Physical	345875	Carla	Basich	857-5566	A528	Lopez
9	12/3/2015	10:30 AM	1.00	Physical	466927	Lisa	Garcia	562-3456	C678	Chapman
10	12/3/2015	9:00 AM	0.50	Migraine	345875	Carla	Basich	857-5666	C678	Chapman

- ❖ It's already in 1NF, and 2NF ...but you need to understand why!
- ❖ **Convert this to 3NF. How many tables do you now have?**

SOLUTION #2

Normalization Practice #2

Appointment Table

<u>Appt No</u>	Appt Date	Appt Time	Planned Duration	Appt Type	Patient ID	First Nm	Last Nm	Phone	Doctor ID	Doctor Nm
1	12/1/2015	3:00 AM	1.00	Physical	466927	Lisa	Garcia	562-3456	C678	Chapman
2	12/1/2015	3:00 AM	0.25	Shot	456789	Sue	Carey	432-1234	A528	Lopez
3	12/1/2015	3:15 AM	0.50	Flu	194756	Brandon	Pierre	432-7877	S626	Smith
4	12/2/2015	10:00 AM	0.50	Migraine	329657	Marcus	Schwartz	239-5502	A528	Lopez
5	12/2/2015	10:15 AM	0.25	Shot	987453	Mike	Jones	456-0202	G123	Gray
6	12/2/2015	10:30 AM	0.25	Shot	384788	Tonya	Johnson	432-8806	S626	Smith
7	12/2/2015	10:45 AM	0.50	Flu	438754	Iliana	Hnatt	823-4303	C678	Chapman
8	12/2/2015	11:00 AM	1.00	Physical	345875	Carla	Basich	857-5566	A528	Lopez
9	12/3/2015	10:30 AM	1.00	Physical	466927	Lisa	Garcia	562-3456	C678	Chapman
10	12/3/2015	9:00 AM	0.50	Migraine	345875	Carla	Basich	857-5666	C678	Chapman

- ❖ It's already in 1NF, and 2NF ...but you need to understand why!
 - 1NF: No Multivalued attributes
 - 2NF: No partial dependencies – entire PK determines each non-key attribute

❖ **Convert this to 3NF. How many tables do you now have?**

Resolving to 3NF

This is the original table

Appointment(ApptNo, ApptDt, ApptTm, PlannedDur, ApptType, PatientID, FirstNm, LastNm, Phone, DoctorID, DoctorNm)

Diagram illustrating the original table structure and dependencies. The table is **Appointment** with attributes: ApptNo, ApptDt, ApptTm, PlannedDur, ApptType, PatientID, FirstNm, LastNm, Phone, DoctorID, DoctorNm. Blue checkmarks are placed above ApptNo, ApptDt, ApptTm, ApptType, PatientID, and DoctorID. Red boxes are placed above PlannedDur, FirstNm, LastNm, Phone, and DoctorNm. Red curved lines indicate dependencies: ApptNo to PlannedDur, ApptNo to FirstNm, ApptNo to LastNm, ApptNo to Phone, and ApptNo to DoctorNm. There are also red boxes below FirstNm, LastNm, and Phone.

3NF SOLUTION:

ApptNo □ ApptDt, ApptTm, PlannedDur, ApptType, PatientID, DoctorID

PatientID □ FirstNm, LastNm, Phone

DoctorID □ DoctorNm

ApptType □ PlannedDur

Normalization Practice #2

Appointment Table

<u>Appt No</u>	Appt Date	Appt Time	Appt Type	Patient ID	Doctor ID
1	12/1/2015	3:00 AM	Physical	466927	C678
2	12/1/2015	3:00 AM	Shot	456789	A528
3	12/1/2015	3:15 AM	Flu	194756	S626
4	12/2/2015	10:00 AM	Migraine	329657	A528
5	12/2/2015	10:15 AM	Shot	987453	G123
6	12/2/2015	10:30 AM	Shot	384788	S626
7	12/2/2015	10:45 AM	Flu	438754	C678
8	12/2/2015	11:00 AM	Physical	345875	A528
9	12/3/2015	10:30 AM	Physical	466927	C678
10	12/3/2015	9:00 AM	Migraine	345875	C678

Patient Table

<u>Patient ID</u>	First Nm	Last Nm	Phone
194756	Brandon	Pierre	432-7877
329657	Marcus	Schwartz	239-5502
345875	Carla	Basich	857-5566
384788	Tonya	Johnson	432-8806
438754	Iliana	Hnatt	823-4303
456789	Sue	Carey	432-1234
466927	Lisa	Garcia	562-3456
987453	Mike	Jones	456-0202

Doctor Table

<u>Doctor ID</u>	Doctor Nm
A528	Lopez
C678	Chapman
G123	Gray
S626	Smith

Appt Type Table

<u>Appt Type</u>	Planned Duration
Flu	0.50
Migraine	0.50
Physical	1.00
Shot	0.25